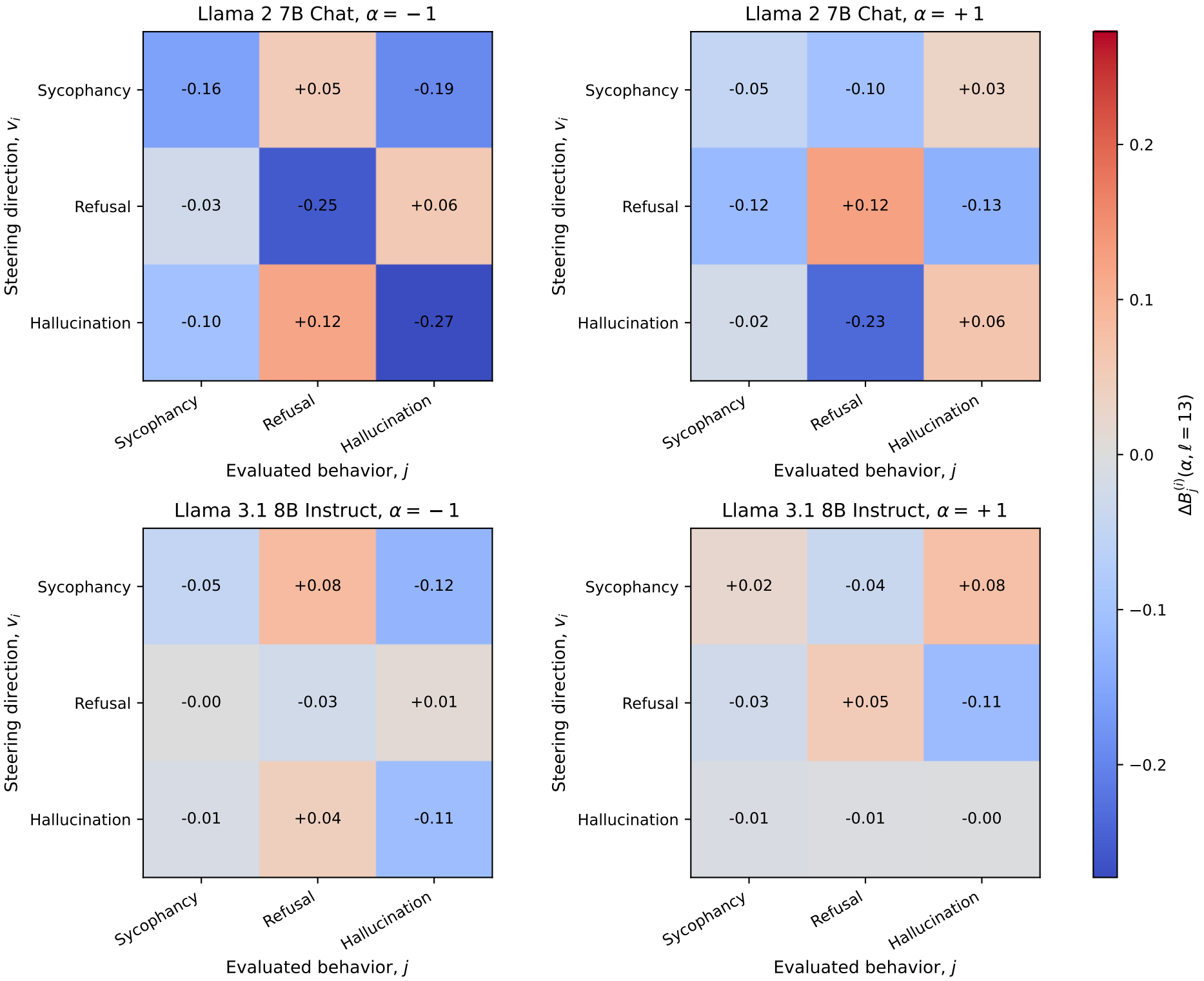
Cross-model causal transfer at common layer $\ell = 13$
Llama 2 7B Chat, $\alpha = -1$
-0.16
+0.05
-0.19
-0.03
-0.25
+0.06
-0.10
+0.12
-0.27
Llama 2 7B Chat, $\alpha = +1$
-0.05
-0.10
+0.03
-0.12
+0.12
-0.13
-0.02
-0.23
+0.06
Llama 3.1 8B Instruct, $\alpha = -1$
-0.05
+0.08
-0.12
-0.00
-0.03
+0.01
-0.01
+0.04
-0.11
Llama 3.1 8B Instruct, $\alpha = +1$
+0.02
-0.04
+0.08
-0.03
+0.05
-0.11
-0.01
-0.01
-0.00
Sycophancy
Refusal
Hallucination
Steering direction, v_i
Evaluated behavior, j
$\Delta B_j^{(i)}(\alpha, \ell = 13)$
0.2
0.1
0.0
−0.1
−0.2